

RoadStar

Dispatch recovery

Carrier decisions. Driver execution.

RoadStar Hackathon 2026



A dock delay reaches the next pickup

A dispatcher must reconnect the next assignment with available resources.

Dock delay

Current truck is still committed

Next pickup

Driver
Truck
Trailer

Illustrative dispatch scenario

Recovery rehearsal

A delayed departure makes the next assignment visible

RS-1042 · Dock departure delayed

Planned end 12:00 p.m. → expected 01:30 p.m. ET

1 Next Assignment At Risk

Planned

Dock delay +90 min

RS-1043 begins 12:15 p.m. ET. Rehearse with a different available driver, truck and trailer.

12:00

Planned departure

13:30

Expected departure

12:15

Next pickup

Synthetic scenario. Times in ET. Actual Ontario truck routing supports the recovery.

Approval against current evidence

The server rechecks the proposal at the moment of approval.

Versions

Availability

Atomic reservations

×

Approve the recovery?

RS-1043 · revision 1

Current assignment

Alex Chen · demo

Truck T-101
Trailer V-101

→

Proposed assignment

Morgan Singh · demo

Truck T-102
Trailer V-102

Driver acceptance is separate

Approval replaces the current assignment and sends a new offer. The proposed driver must accept it.

► Checks & assumptions

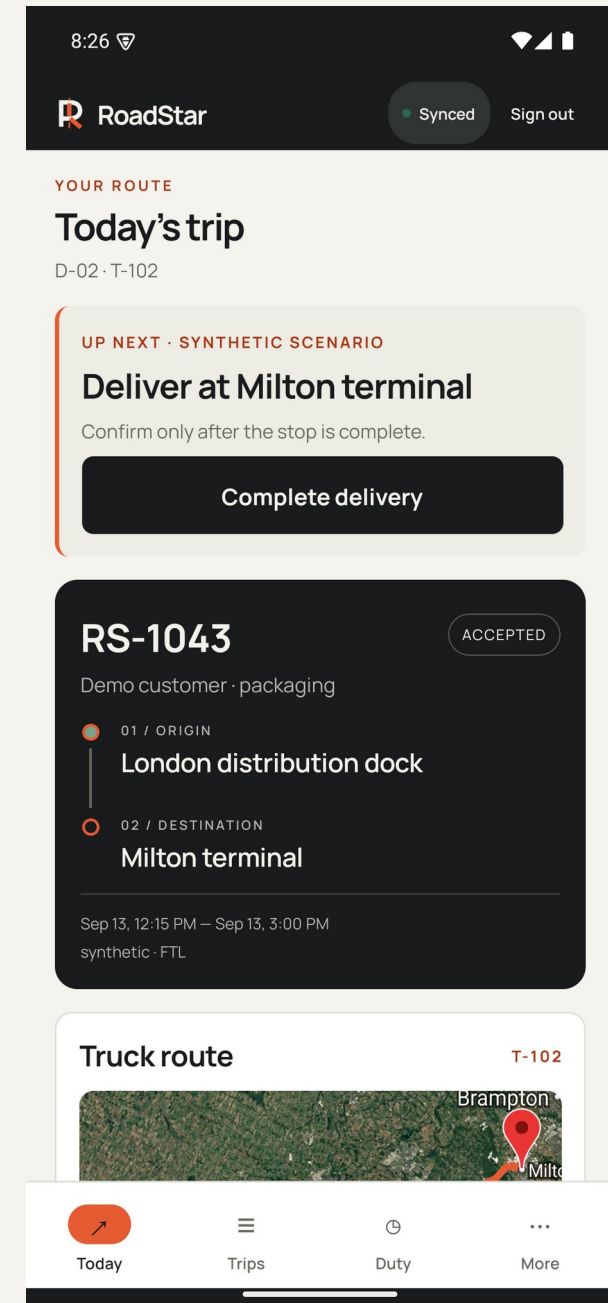
Approve revision 1

The driver keeps the work

Acceptance stays visible through an offline interruption.

- 01 Pending while offline
- 02 Trip and draft survive termination
- 03 Server confirms the command

Android emulator verified. Physical phone and native iOS remain unverified.



Documents and detention

A reviewed revision keeps the original draft in view

RS-1042 · milton-yard Sep 13, 2026, 8:30 a.m. → Sep 13, 2026, 11:17 a.m.		Review stop evidence
RS-1042 · revision 2 167 minutes observed · 47 billable minutes Timestamp precision: observed samples ► Approval evidence		\$78.33 Approved
RS-1042 · revision 1 167 minutes observed · 47 billable minutes Timestamp precision: observed samples · historical revision; latest is 2		\$78.33 Draft

Observed samples require review. Billing approval does not initiate payment.

2 loads

4 ordered stops

One vehicle pairing
One approval

Reservations stay active until every load completes.

Synthetic pallet capacity disclosed

Shared route · one manifest

Completed

2 loads · 4 ordered stops

D-01 · Truck T-101 · Trailer V-101

synthetic · valhalla-truck · 98 modeled driving minutes · 138 modeled duty minutes

All stops are completed. This manifest remains available for reference.

1. RS-1042 · pickup
Milton terminal · demo location
Completed

2. RS-1044 · pickup
Milton terminal · demo location
Completed

3. RS-1042 · delivery
London distribution dock · demo location
Completed

4. RS-1044 · delivery
London distribution dock · demo location
Completed

► Planning assumptions

All five sheets retain lineage

15,197

source rows

82 exact Dispatch duplicates

Historical provenance preserved
Private source data stays local

Sheet	Rows
Tlorder	4,031
Dispatch	10,479
Driver	131
Trucks	131
Trailers	425

One operational database

CLIENTS

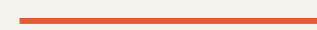
COMMANDS

STORAGE

Web + mobile



Fastify API



PostgreSQL

React dispatcher UI
Expo driver execution

Firebase identity
Carrier membership
Validated commands

Versions
Reservations
Evidence

Supporting capabilities used by the API

Proposals: OR-Tools + Valhalla

Documents: Gemini extraction

Independent replay clock · Approval remains an explicit command.

Measured proof

5.4 s

p95 acknowledgement-to-snapshot lag
in the 131-driver cloud burst

22 backend checks
23 mobile checks
Real-road integration verified separately

524

telemetry + synchronization
pairs
Zero failed commands

Physical background location
and native iOS still need testing.
Declared duty budgets are not a
certified ELD.

Next carrier evaluation

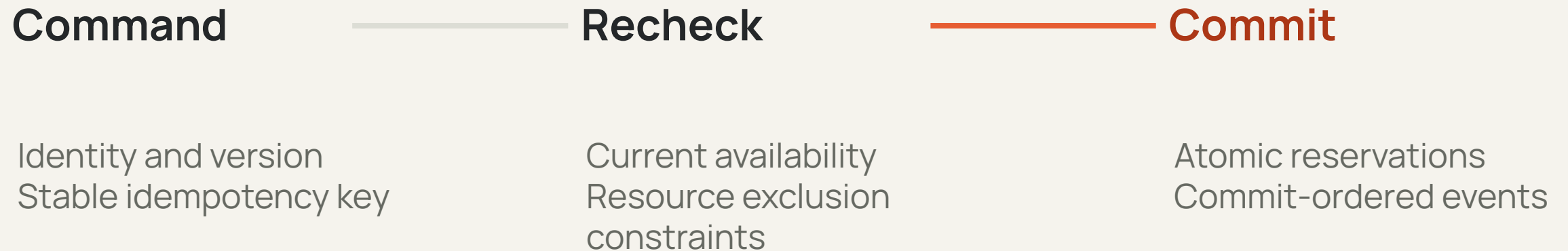
Physical device validation
Carrier inputs and facility review
Measured dispatcher and driver use

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Concurrent approval

A stable command key and expected version protect the same intent



A stale proposal conflicts before mutation. A repeated key returns its original result.

Replay and telemetry evidence

Replay conditions

- Recorded starting conditions
- Disclosed dock wait and road assumptions
- An independent scenario clock

Telemetry interpretation

- Duplicate samples do not double-count
- Out-of-order samples retain history
- Uncertain boundaries stay uncertain

Operational outcomes remain separate from modeled savings.